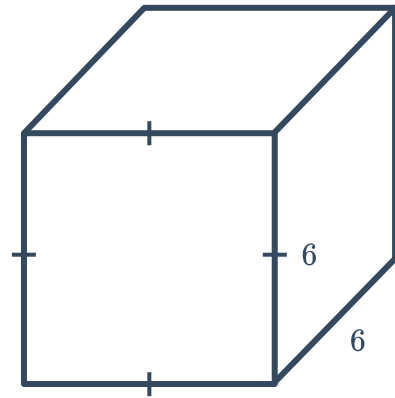


Surface area of prisms



- 1 Consider the following cube with a side length of 6 cm:

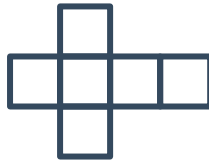


- a Which of the following nets match the given cube?

A



B



C

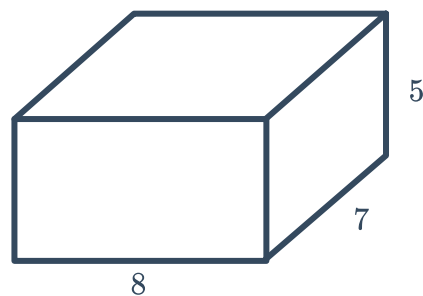


D



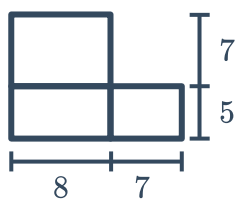
- b Find the area of the cube's net.
- c Find the surface area of the cube.

2 Consider the following rectangular prism:

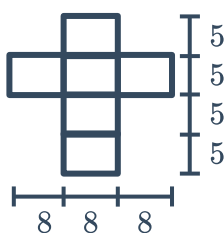


a Which of the following nets match the given rectangular prism?

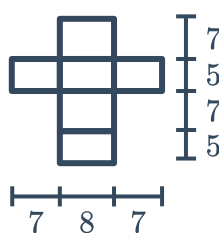
A



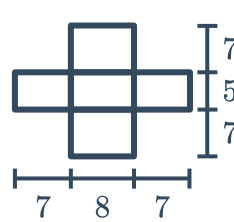
B



C



D

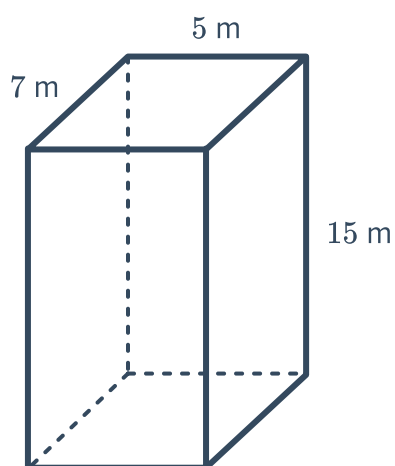


b Find the area of the prism's net.

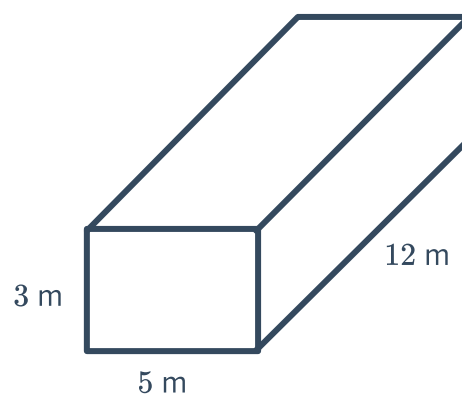
c Find the surface area of the rectangular prism.

3 Find the surface area of the following rectangular prisms:

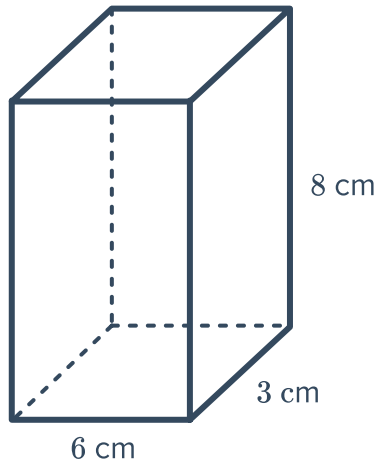
a



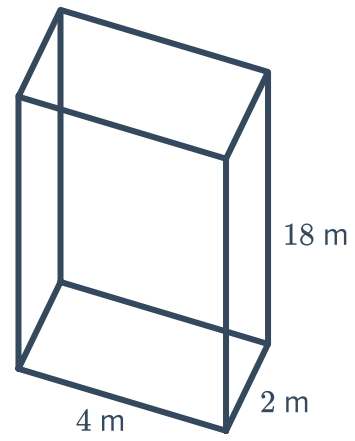
b



c

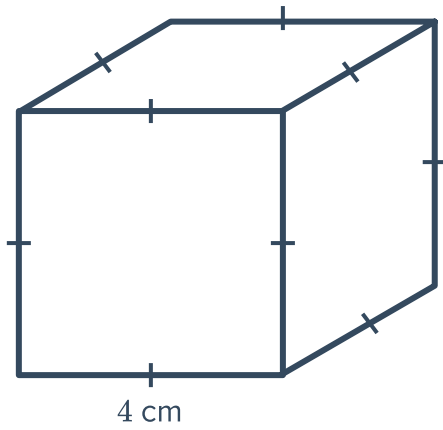


d

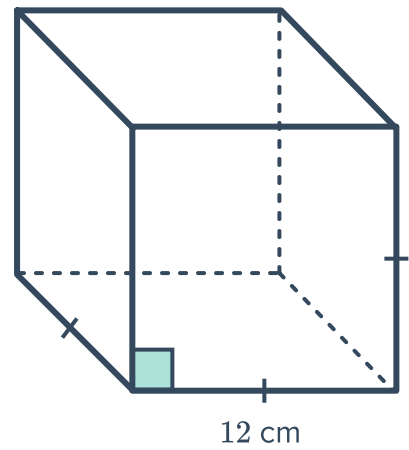


4 Find the surface area of the following cubes:

a



b

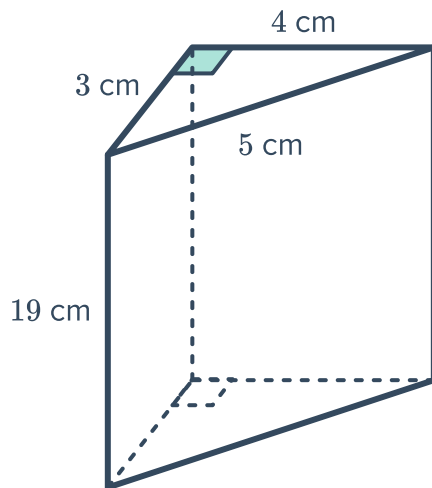


5 Find the side length of a cube that has a surface area of 384 cm^2 .

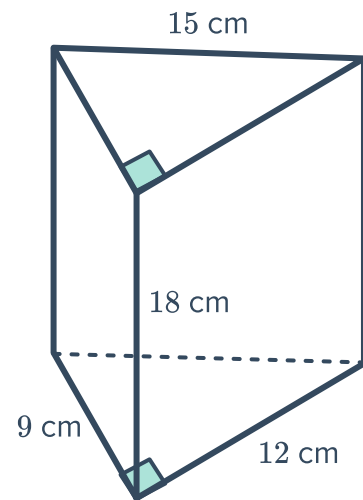
6 Find the surface area of the following triangular prisms:



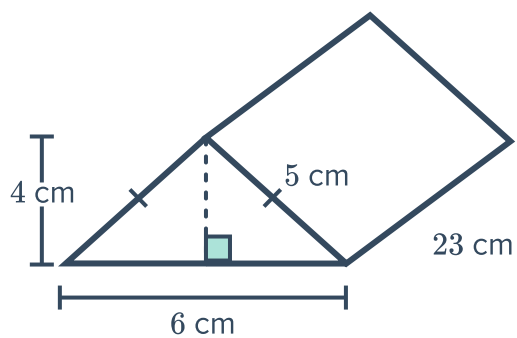
a



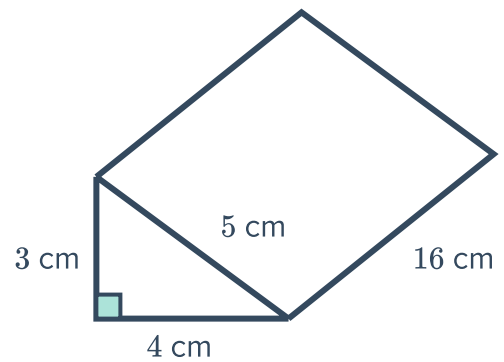
b



c

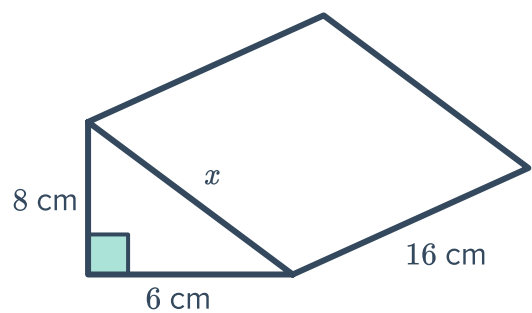


d



7 Consider the following figure:

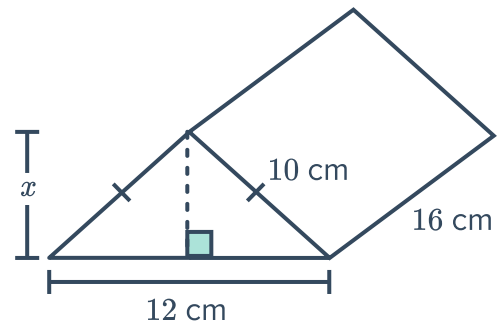
- a Find the value of x .
- b Find its surface area.



8 Consider the following figure:

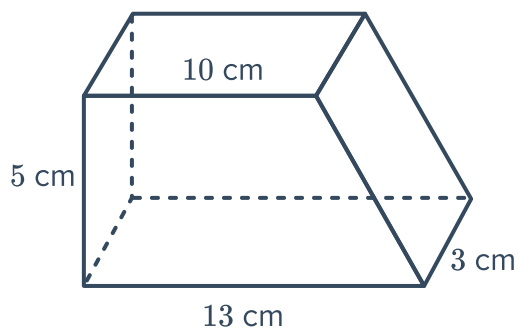


- a Find the value of x .
- b Find its surface area.

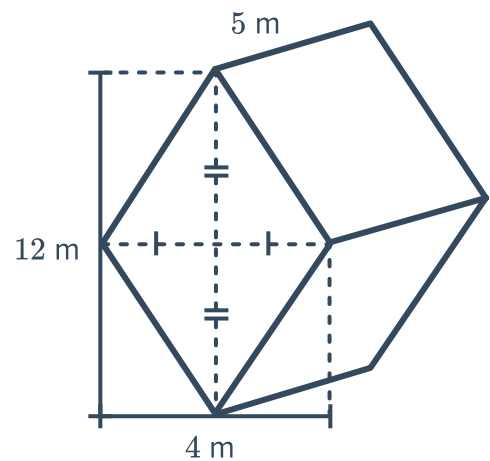


9 Find the surface area of the following prisms. Round your answer to two decimal places when necessary.

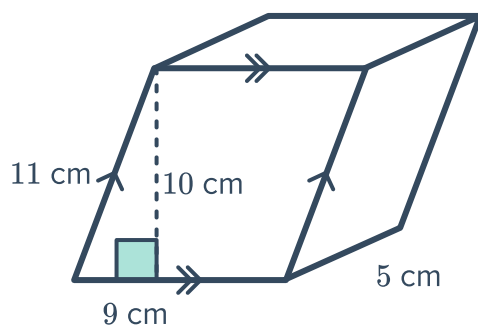
a



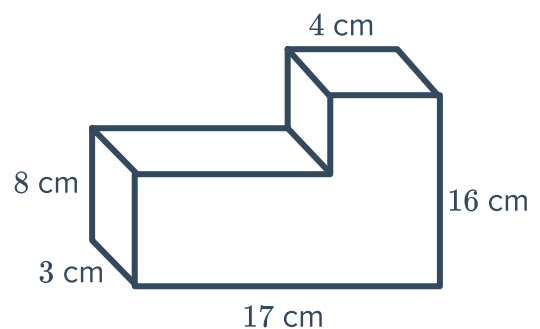
b



c



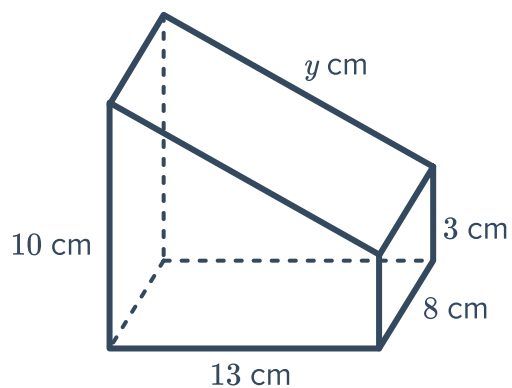
d



10 Consider the following trapezoidal prism:

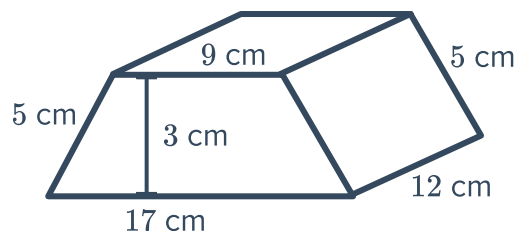


- a Find the value of y correct to one decimal place.
- b Find its surface area.



11 Consider the following trapezoidal prism:

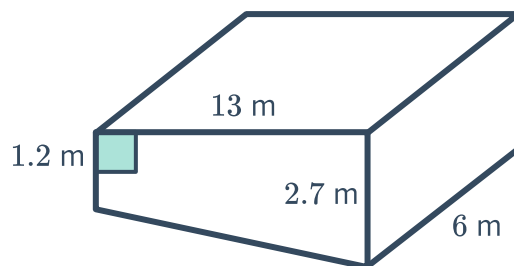
Find its surface area.



Applications

12 Consider the swimming pool with the following dimensions:

Calculate the area inside the pool that is to be tiled. Round your answer to two decimal places.



- 13 Sally is building a storage chest in the shape of a rectangular prism. The chest will be 93 cm long, 80 cm deep, and 18 cm high. Find the surface area of the chest.

- 14 A birthday gift is placed inside the box shown:

Find the minimum amount of wrapping paper needed to wrap this gift?



- 15 Both of the following popcorn bags are designed to carry 735 cm^3 of popcorn.

- a Find the height h of bag A.
- b Find the height h of bag B.
- c Find the surface area of bag A.
- d Find the surface area of bag B.
- e Which bag should be used to reduce the amount of paper needed?

